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CHM 311 — Past Questions & Materials
by Renance

Course: CHM 311
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 300

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Paper 1 of 2 — CHM 311 - Transition Metals Chemistry

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE, DEPARTMENT OF CHEMISTRY
2024/2025 (FIRST SEMESTER) EXAMINATIONS
CHM 311 (Transition Metal Chemistry)

Instruction: Answer FOUR questions in all (At least ONE questions from each section)

Time allowed: 2½ hrs

A

1. (a) What are the transition elements, why are they called so? (5½ marks)
(b) Mention 3 factors that enable Transition Elements to form Complexes. (5 marks)
(c) Write down the names and symbols of 4d and 5d congeners of element Manganese. (5 marks)
(d) Which ion is more stable Ag^+ or Ag^{2+} and why? (2 marks)
2. (a) Explain the origin of ligand field splitting in transition metal complexes. How does it relate to the crystal field theory? (6½ marks)
(b) Discuss the factors that influence the magnitude of ligand field splitting in a transition metal complex. (6 marks)
(c) State the difference between weak-field and strong-field ligands. How do they affect the electronic structure and magnetic properties of transition metal complexes? (5 marks)

B

3. (a) Why do Cr and Cu have anomalous configurations? (4 marks)
(b) The melting points of first transition series metals increase, reach a maximum and then decrease. Give reasons. (5 marks)
(c) Explain the following variations in terms of electronic configurations: The second ionization enthalpies of both Cr and Cu are higher than those of their next elements respectively. (5 marks)
(d) Give reasons for the variable oxidation states found in d-block elements. (4 marks)
4. (a) Why do the elements of second and third transition series show strong similarity in the chemistry within a group but show large differences from the elements of the first transition series? (5 marks)
(b) Write electronic configurations for Mo, Rh and Os [$^{96}_{42}\text{Mo}$, $^{103}_{45}\text{Rh}$, $^{190}_{76}\text{Os}$] (4 marks)
(c) Enumerate the factors which determine the stability of different oxidation states of transition metals. (5 marks)
(d) The most stable oxidation state of Zr is ZrCl_4 , while that of Pd is PdCl_2 . Why? (4 marks)

C

5. (a) Is lanthanum ($_{57}\text{La}$) really a lanthanoid element (Yes/No)? Explain. (3 marks)
- (b) Lanthanides most notably occur in only one oxidation state.
- (i) What is this oxidation state; and what is its origin? (4 marks)
 - (ii) Name two other oxidation states in which lanthanoids can occur (2 marks)
 - (iii) Explain briefly the origin of these two oxidation states. (2 marks)
 - (iv) Name two lanthanoid elements (one in each case) that can occur in the oxidation states mentioned in b(ii) above. (2 marks)
- (c) What do gadolinium ($_{64}\text{Gd}$) and lutetium ($_{71}\text{Lu}$) have in common in their electron configurations? Comment briefly on this. (4 marks)
- (d) The actinoid elements are paramagnetic; true or false? (1 mark)
6. (a) (i) What is "lanthanoid contraction"? (2 marks)
- (ii) Explain briefly the cause of lanthanoid contraction. (4 marks)
 - (iii) Highlight three consequences of the lanthanoid contraction. (4 marks)
- (b) (i) Is Samarium a lanthanoid or an actinoid element? (1 mark)
- (ii) Write the electron configuration for samarium ($_{62}\text{Sm}$). (2 marks)
- (c) In tabular form, highlight three differences between lanthanoids and actinoids. (3 marks)
- (d) Give two uses (each) of lanthanoids and actinoids. (2 marks)

Paper 2 of 2 — CHM 311 Transition metal chemistry 2024/2025

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF SCIENCE, DEPARTMENT OF CHEMISTRY

2024/2025 (FIRST SEMESTER) EXAMINATIONS

CHM 411 (Transition Metal Chemistry)

on: Answer FOUR questions in all (At least ONE questions from each section)

Time allowed: 2 hrs

Instructions

1. What are the transition elements, why are they called so? (5 marks)

2.

(9) Mention 3 factors that enable Transition Elements to form Complexes [> 7 ans]

3. (5

(c) Write down the names and symbols of 4d and 5d congeners of tungsten and

4. (5

5. Which ion is more stable Ag^+ or Ag ? and why? (

6. (2

(a) Explain the origin of ligand field splitting in transition metal complexes. How does it relate to the crystal field theory? (6% marks)

7. In d-d transition,

(5) Discuss the factors that influence the magnitude of ligand field splitting

8. (6 marks)

9. Explain the difference between weak-field and strong-field ligands. How does it affect

10. (5

11. Explain the electronic structure and magnetic properties of transition

12. (5

13. Why do Cr and Cu have anomalous configurations? (4 marks)

14. (5

15. (5

16. The melting points of first transition series metals increase, reach a maximum and then decrease. Give reasons. (5 marks)

17. Explain the following variations in terms of electronic configuration: The second

18. Ionization enthalpies of both Cr and Cu are higher than those of their next elements

19. respectively. (5 marks)

20. (5

21. Give reasons for the variable oxidation states found in d-block elements. (4 marks)

fa} Why do the elements of second and third transition series show strong similarity '9 the + of all 4

chemistry within a group but show large differences from the elements of the first transition series? (5 marks)

'hb' Write electronic configurations for Mo, Rh and Os [38Mo, +9 19

ip atomic configurations for Mo, Rh and Os [42Mo, *G2Rh, 4220s} (4 marks)

(ci Enumerate the factors which determine the stability of different oxidation states of OF - a 19 ates &

transition metals. (5 marks)

(d) The most stable oxidation state of Zr is 2rCl₄, while that of Pd is PdCl. Why? (4 marks:

iv :

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Cc 5. (a) Is lanthanum (.7L4} really a lanthanoid element (Yes/No)? Explain. (2 marks)

(aes (o) Lanthanides most notably occur in only one oxidation state.

(i) What is this oxidation state; and what is its origin? (4 marks)

(ii) Name two other oxidation states in which lanthanoids can occur! (2 marks)

(iii) Explain briefly the origin of these two oxidation states. (2 marks:

(iv) Name two lanthanoid elements (one in each: Case) that can occur at the oxidation states mentioned in b(ii) above. {2 marks)

{c} What do gadolinium (ssGd) and lutetium (-.Lu) have in common in their electron configurations? Comment briefly on this. (4 marks)

{a} The actinoid elements are paramagnetic; true or false? (4 mark)

fia (4 What is "Lanthanoid contraction"? (2 marks)

(i) Explain briefly the cause of lanthanoid contraction. (4 marks)

Tell me 2 lanthanoid or an actinoid element? (1 mark) a

Co Write the electron configuration for samarium (Sm), (2 marks)

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